Pulmonary artery pressure waveform analysis in predicting heart failure hospitalization

A secondary analysis of the PROACTIVE-HF trial

Liviu Klein, MD, MS

On behalf of the PROACTIVE-HF investigators

CRF

THT

TECHNOLOGY AND HEART FAILURE THERAPEUTICS

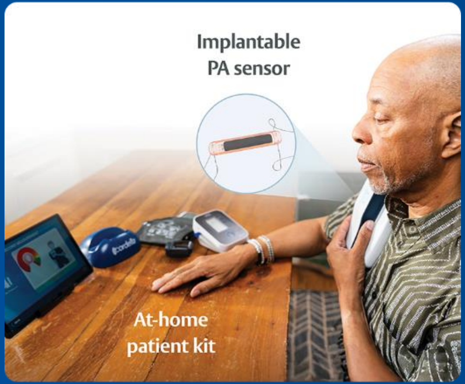
Disclosure of Relevant Financial Relationships

Within the past 24 months, I or my spouse/partner have had a financial interest/ arrangement or affiliation with the following ineligible companies.

Affiliation/Financial Relationship	Company
Grant/Research Support	Ancora Heart, Alleviant, Axon Therapies, Cordio, Endotronix, Edwards, Impulse Dynamics, Paragonix, Procyrion, Revamp Medical, TransMedics, V-wave Medical
Consulting Fees/Honoraria	Abbott, Adona Medical, Edwards, Supira Medical
Major Stock Shareholder/Equity	Cardiosense
Royalty Income	N/A
Ownership/Founder	N/A
Intellectual Property Rights	N/A
Other Financial Benefit	N/A

CRF THT

Background



Remote management of pulmonary artery pressure (PAP) and vital signs result in low rates of heart failure hospitalizations (HFHs)

There remains some residual HFHs when following the current management protocol

PAP waveforms (WVFs) may encode additional, actionable information beyond pressure magnitude that could assist in determining patients at risk of residual HFHs

Objective

Goal: ***Develop a machine learning (ML) model to predict the number of days to first HFHs using PAP waveforms***

Methods: Data

1 Year Proactive HF Study Data

- 18-second waveform data from Cordella PA pressure sensor for patients in seated position taken daily
- Waveforms recorded at 814 Hz yielding approximately 15,000 datapoints per waveform
- Blood oxygen (SpO2), blood pressure, weight, and heart rate measured by the Cordella system was used

Methods: Analysis

Model Inputs

Waveform Features:
45 days prior to first HFHs

Baseline Features:
demographics, medical history; included in waveform & reference models

Vital sign Features:
sBP, weight, HR; included in waveform & reference models

Mean PAP:
Included in reference model only

Algorithms

- **Random Forest**
- **Gradient Boost**
- **Elastic Net**
- **Support Vector Regression**

Cross-Validation Strategy

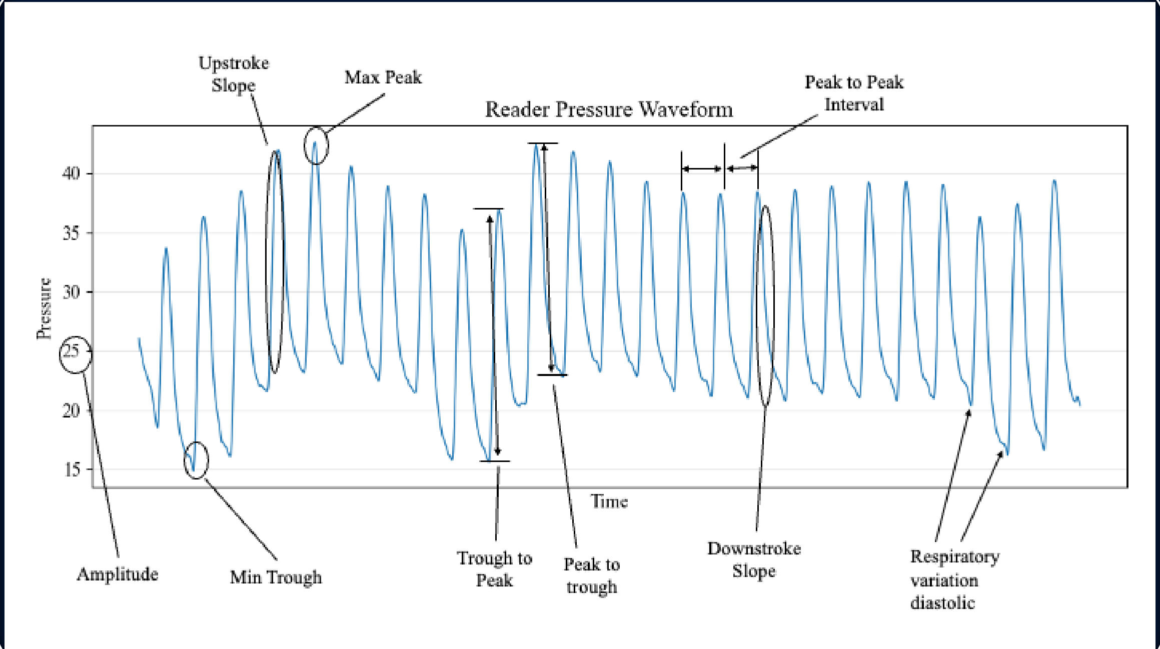
5-fold cross validation with an 80% / 20% split for training and testing dataset along with 20% split for validation dataset

Metrics

- **R^2**
- **Mean Absolute Error (MAE)**

Note: ML waveform model's accuracy was compared to reference model trained with baseline data, daily vital sign data, and daily mPAP

Pulmonary Artery Waveform Features



Types of Features

26 Basic Features

Highlight important physiological elements

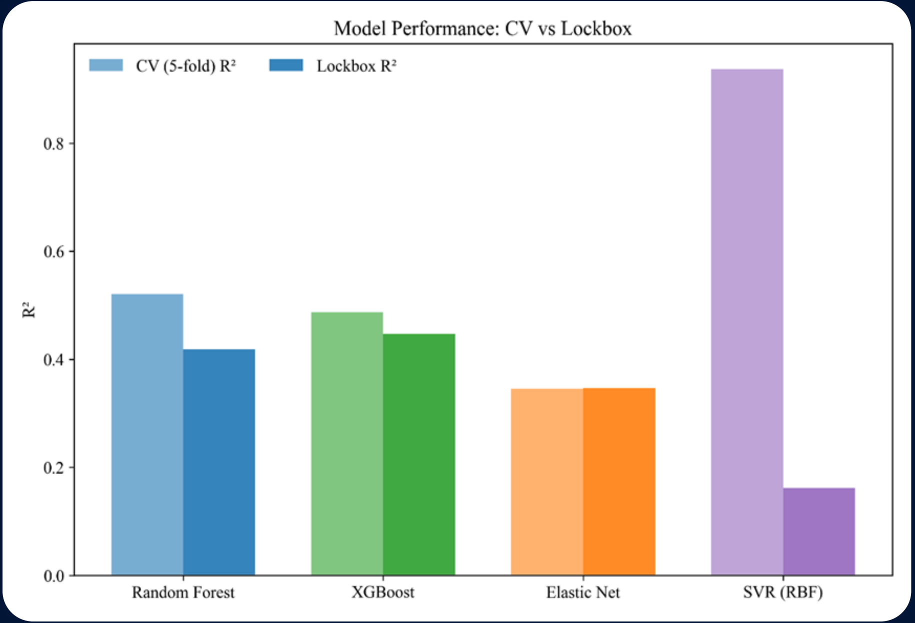
Enhanced Temporal Features

Captures baseline-deviation, relative standing among prior waveforms, and trends over time

Interaction Features

Captures the relationship among multiple features

Algorithm Comparison



Takeaways

- **XGBoost performed the best in the lockbox test dataset**
  - ***R^2:* The model explains 44% of the variability in time to HFH events**
  - ***MAE:* The model has an average prediction error of 7.5 days**
- **In multiple versions, SVR struggled with generalizability (likely due to overfitting)**

***Clinical Implication*: the model produces meaningful early warning by estimating proximity to HFH within a week**

Reference Model Features

Baseline Charecteristics

Variable	Baseline Values
LV EF %	39 ± 18 %
Sex	M (64%) or F (36%)
Age	63 ± 12
BMI	34 ± 8 kg^2/m^2
HF type	Ischemic (25%)
# hospitalizations year before implant	1.5 ± 1

Demographic data was recorded at the beginning of the study

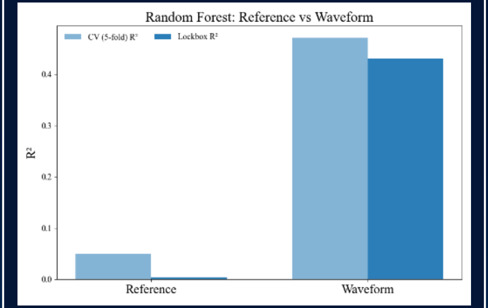
Daily mPAP and Vital Signs

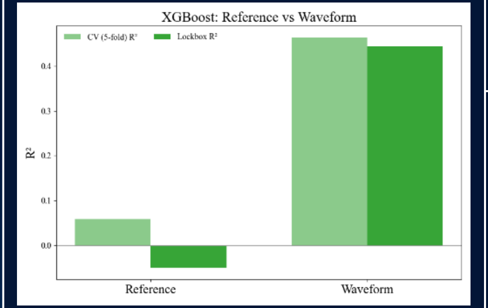
Variable	Values Collected Over Analysis Period
mPAP	27 ± 12 mmHg
Systolic BP	116 ± 20 mmHg
Diastolic BP	69 ± 12 mmHg
SpO2	95 ± 3 %
Weight	224 ± 56 lbs
Pulse	79 ± 14 bpm

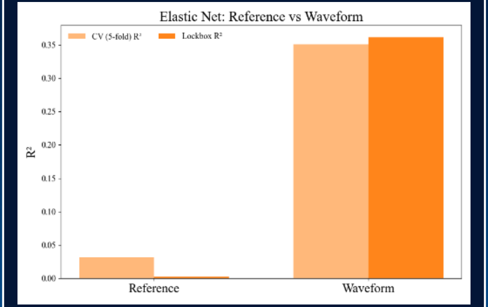
Vital signs were collected daily through PROACTIVE HF Study

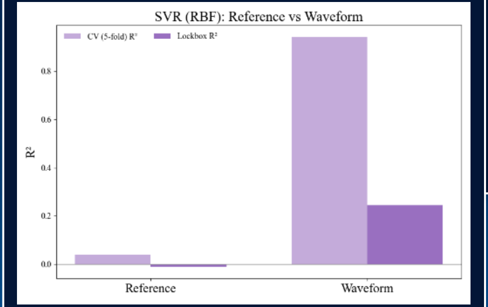
Reference model was created using these features with the same approach as the waveform model

Comparison Against Reference Model









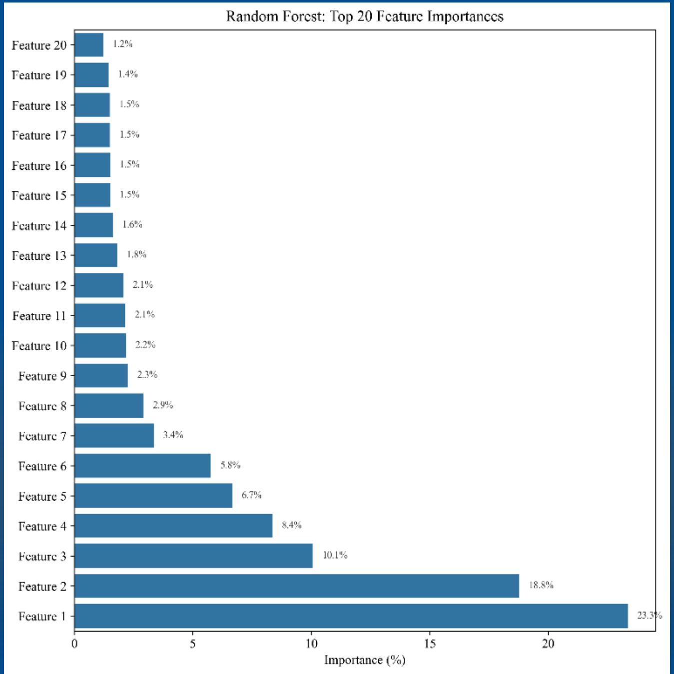
Takeaways

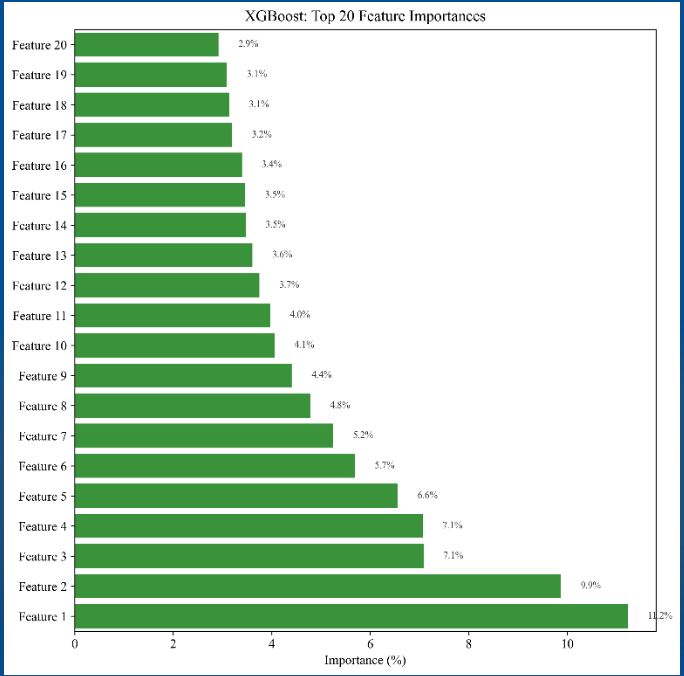
The waveform model significantly outperforms the reference model

- **<u>The waveform model</u> shows a strong predictive signal capturing meaningful physiological patterns**
- **<u>The reference model</u> performs *no better* than predicting the average time-to-event for patients**

***<u>Clinical Implication</u>*: Waveform derived inputs substantial improve the model's ability to estimate proximity to HFH**

Feature Importances Distribution

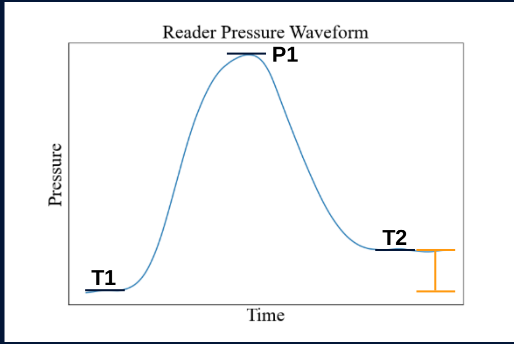




Random Forest and XGBoost have the same top 3 dominating features

Top Features

Systolic to Diastolic Ratio – Subject Percentile

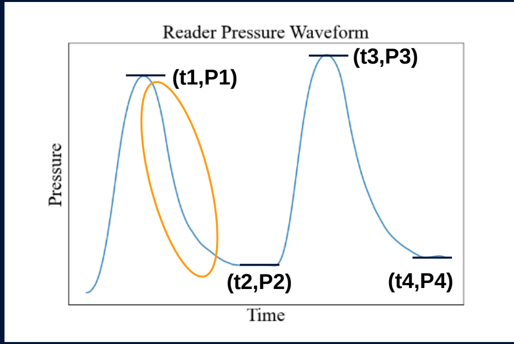


$$Ratio = \frac{systolic}{diastolic} = \frac{P1 - T1}{T2 - P1}$$

Clinical Implication:
Where does the ratio of systolic duration to diastolic duration (sys/dia) for this waveform lie as a percentile of all other measurements prior to this reading?

A higher sys/dia ratio than usual indicates increased risk

Minimum Downstroke Slope vs. Personal Mean

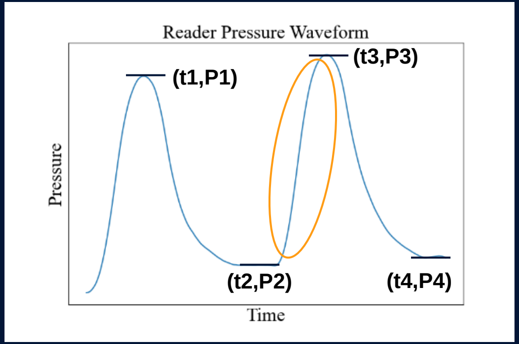


$$downstroke\ slope = \frac{P1 - P2}{t1 - t2}$$

Clinical Implication:
How does the minimum downstroke slope from the current waveform compare to the patient's mean of minimum downstroke slopes across all readings prior?

A less steep than usual most extreme downstroke slope indicates increased risk

Max Upstroke Slope – Subject Percentile



$$upstroke\ slope = \frac{P3 - P2}{t3 - t2}$$

Clinical Implication:
Where does the maximum upstroke slope in the current waveform lie as a percentile of all other upstroke slope measurements prior to this reading?

A steeper than usual most extreme upstroke slope indicates increased risk

Minimal collinearity among top features (all feature pairs have Spearman correlation < .2)

Conclusion

- Time to HFH is *predictable* using PA pressure waveforms
- Time to HFH is *not predictable* using the current methods from mean PA pressure alone
- This suggests PA pressure waveforms contain more useful information for understanding the risk of HFH
- The most important features in predicting HFH risk appear to be:
  - Systolic to Diastolic Ratio – Subject Percentile
  - Minimum Downstroke Slope vs. Personal Mean
  - Max Upstroke Slope – Subject Percentile
- There are numerous avenues for improving the accuracy of these predictions in future studies